

Instructions



**Tell me if you have problems with
practictical tasks**

How is it???



good

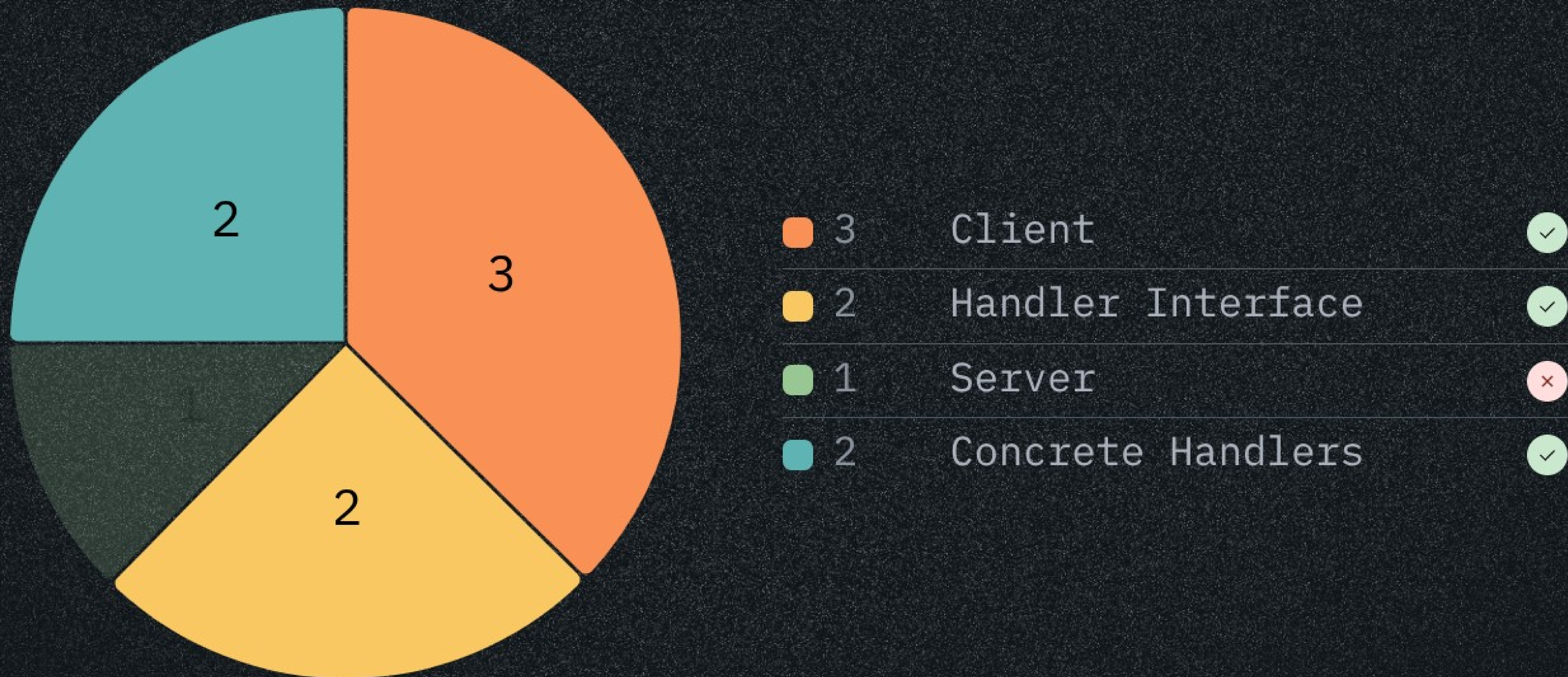
Agenda

- Recall what was learned
- Look at current topic practice
- Discuss theory
- Quiz

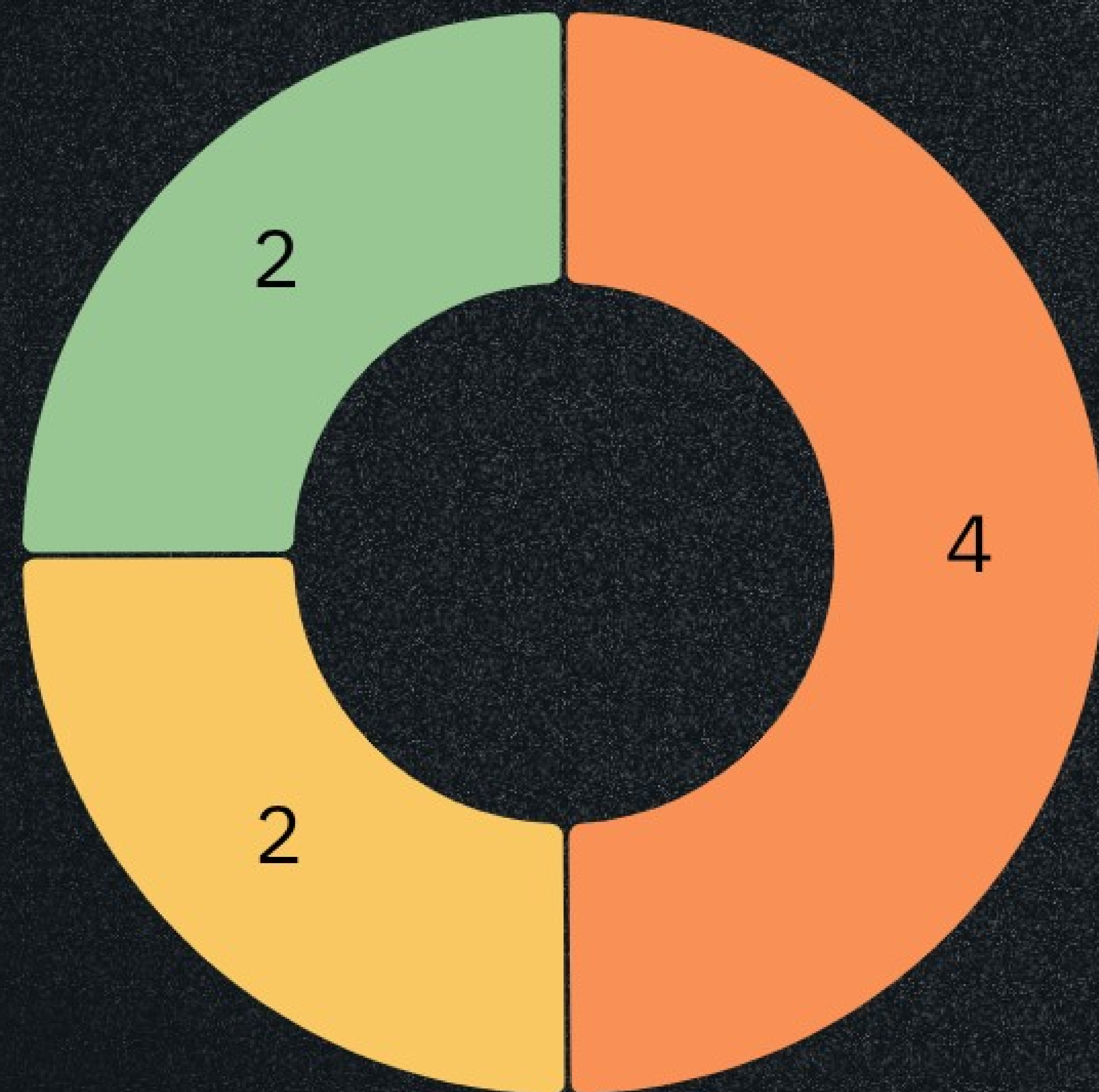
Total Recall



What are the key components of the chain of responsibility pattern?

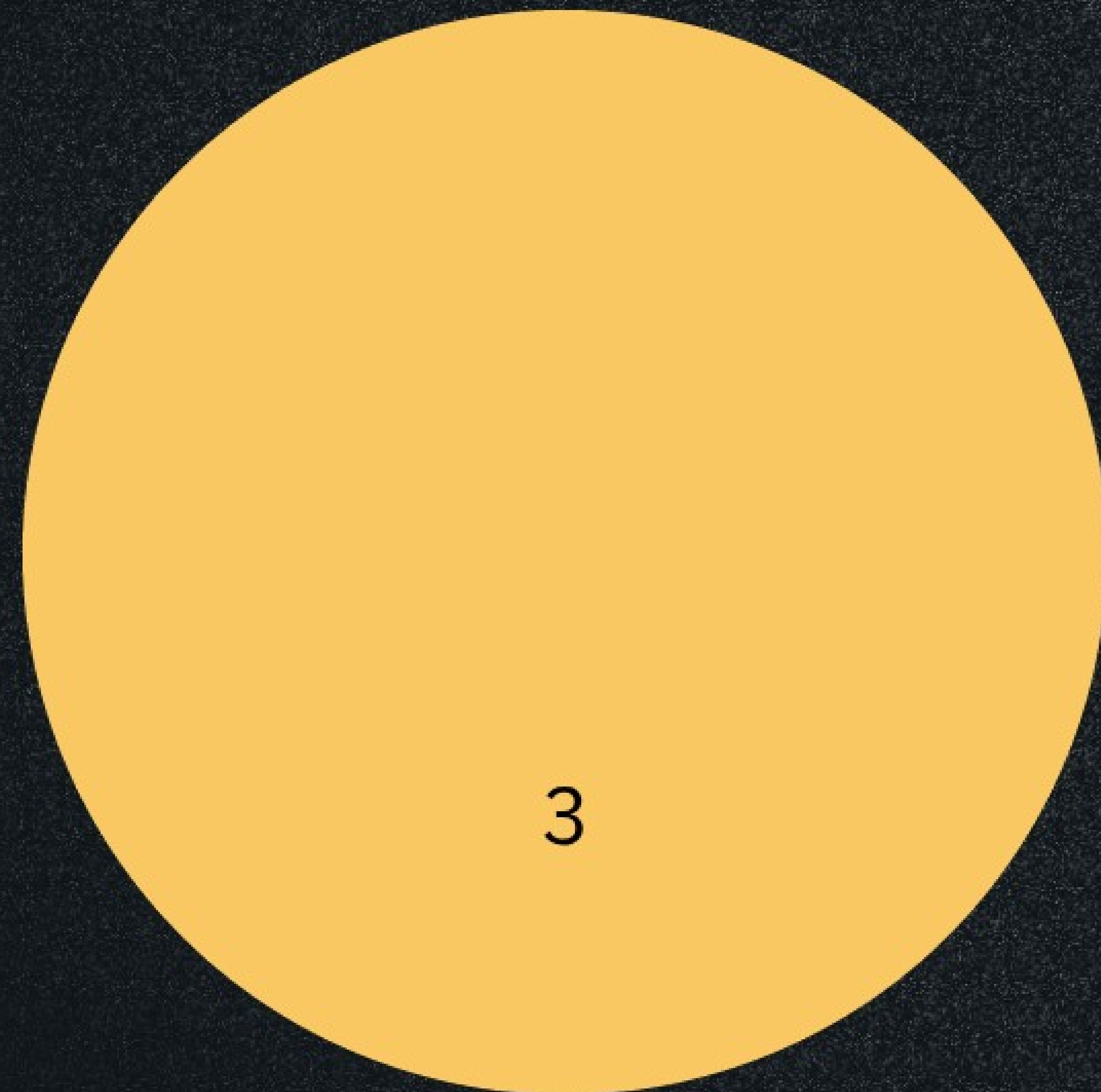


Select all usage scenarios in which Chain of Responsibility can be particularly beneficial:



4	Event Handling in UI Frameworks	✓
2	Approval Processes	✓
2	Command Processing	✓
0	Memory Optimization	✗

In the Chain of Responsibility, if an object can handle a request, what should it do next?



- | | | |
|---|---|---|
| 0 | A) Pass the request to all successor handlers irrespective of its own ability to handle it. | ✗ |
| 3 | B) Handle the request and decide whether to pass it to the next handler in the chain. | ✓ |
| 0 | C) Modify the request and send it back to the sender. | ✗ |
| 0 | D) Terminate the program. | ✗ |

Discuss current topic

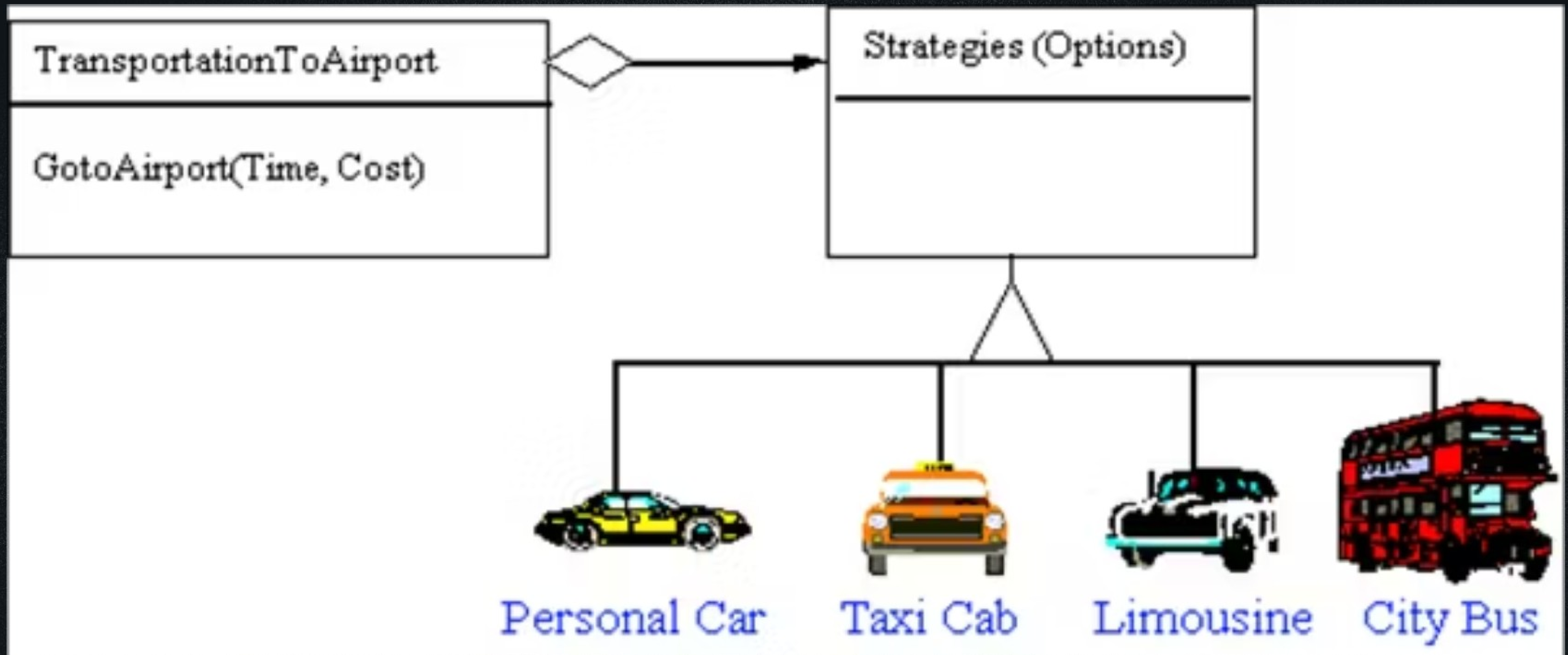


Over 90% of the world's currency is digital.

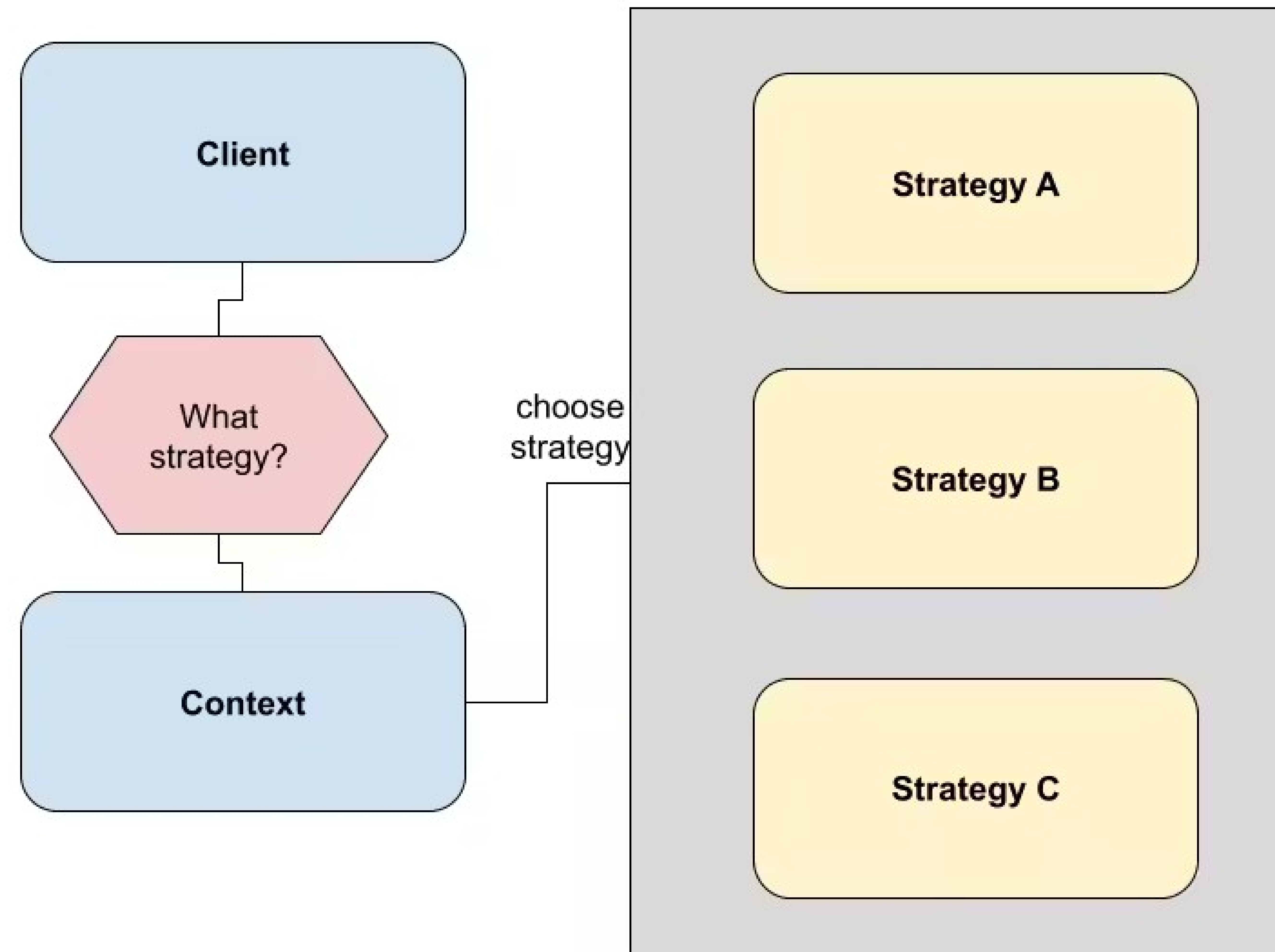
Between credit cards, debit cards, online purchases, and cryptocurrency, only 8% of global currency is physical money.

Have you looked through the materials?

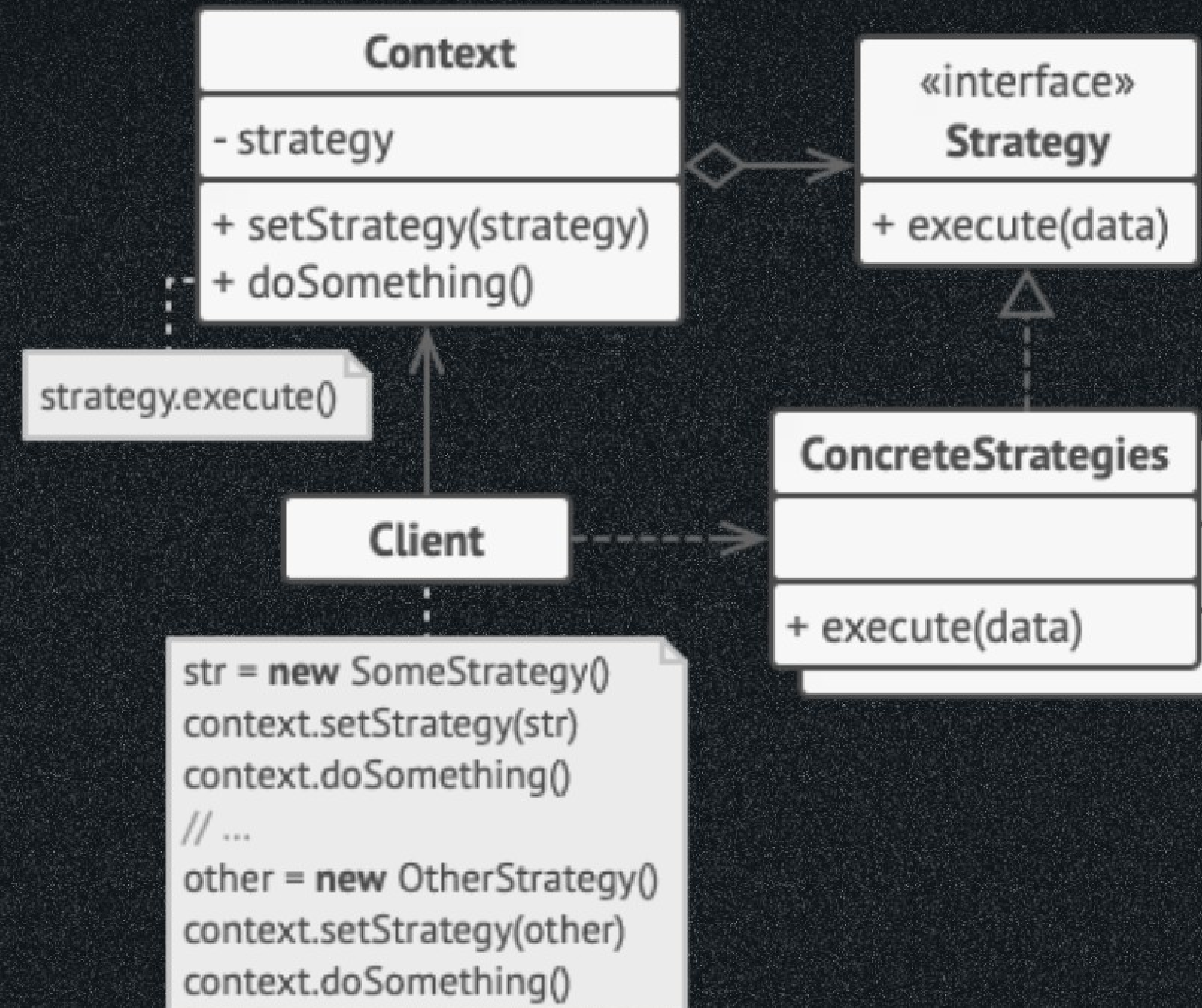




Descriptive real-world example



Advanced Patterns: Strategy



Original structure of Strategy

Considerations and Best Practices

- Consistent interfaces: Ensure all strategies implement a consistent interface to facilitate interchangeability.
- Document: Clearly document each strategy and when it should be used.
- Moderate use: Avoid creating excessive unnecessary strategies that complicate the system.

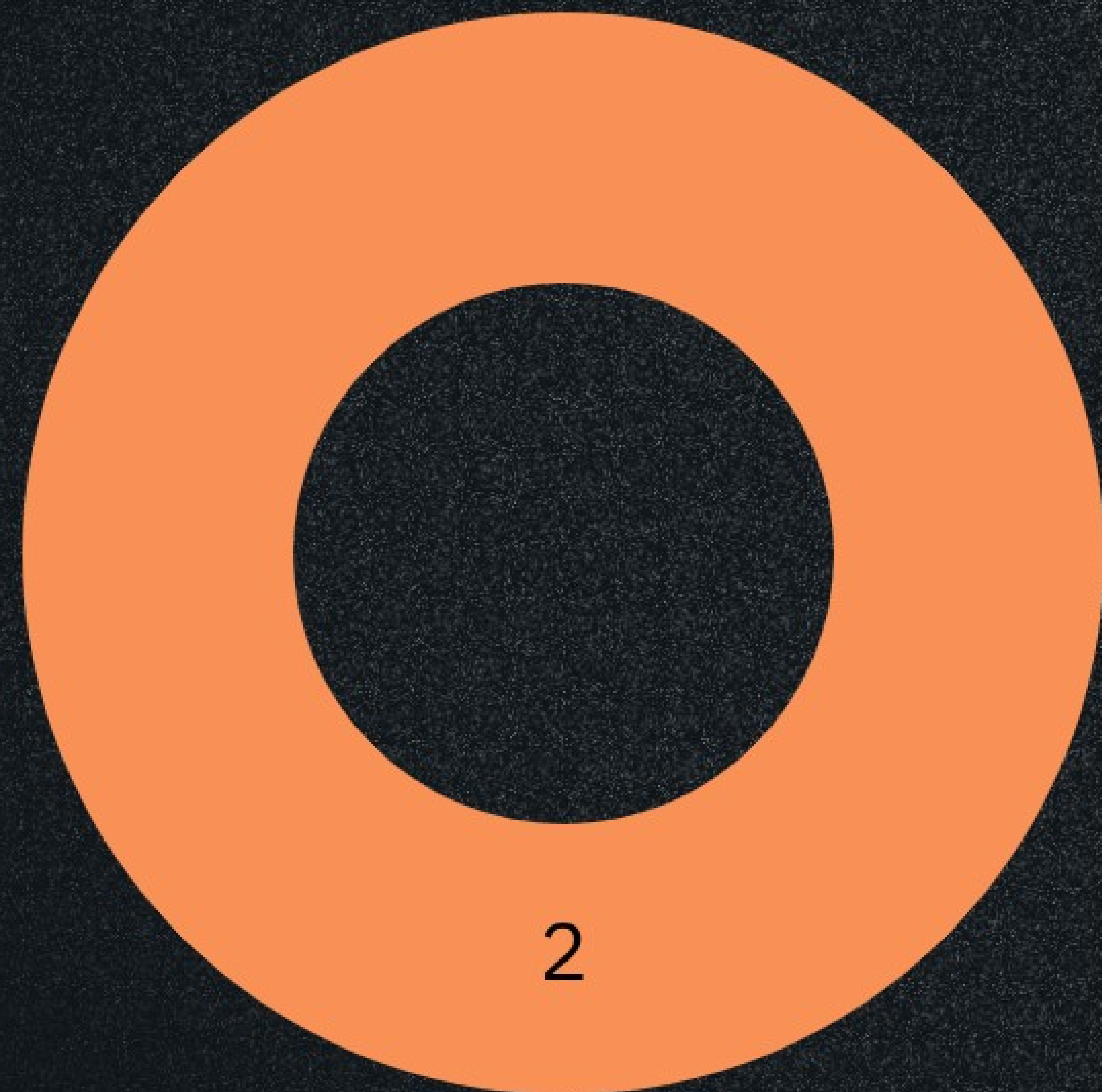
Benefits

- You can swap algorithms used inside an object at runtime.
- You can isolate the implementation details of an algorithm from the code that uses it.
- You can replace inheritance with composition.
- Open/Closed Principle. You can introduce new strategies without having to change the context.

Disadvantages

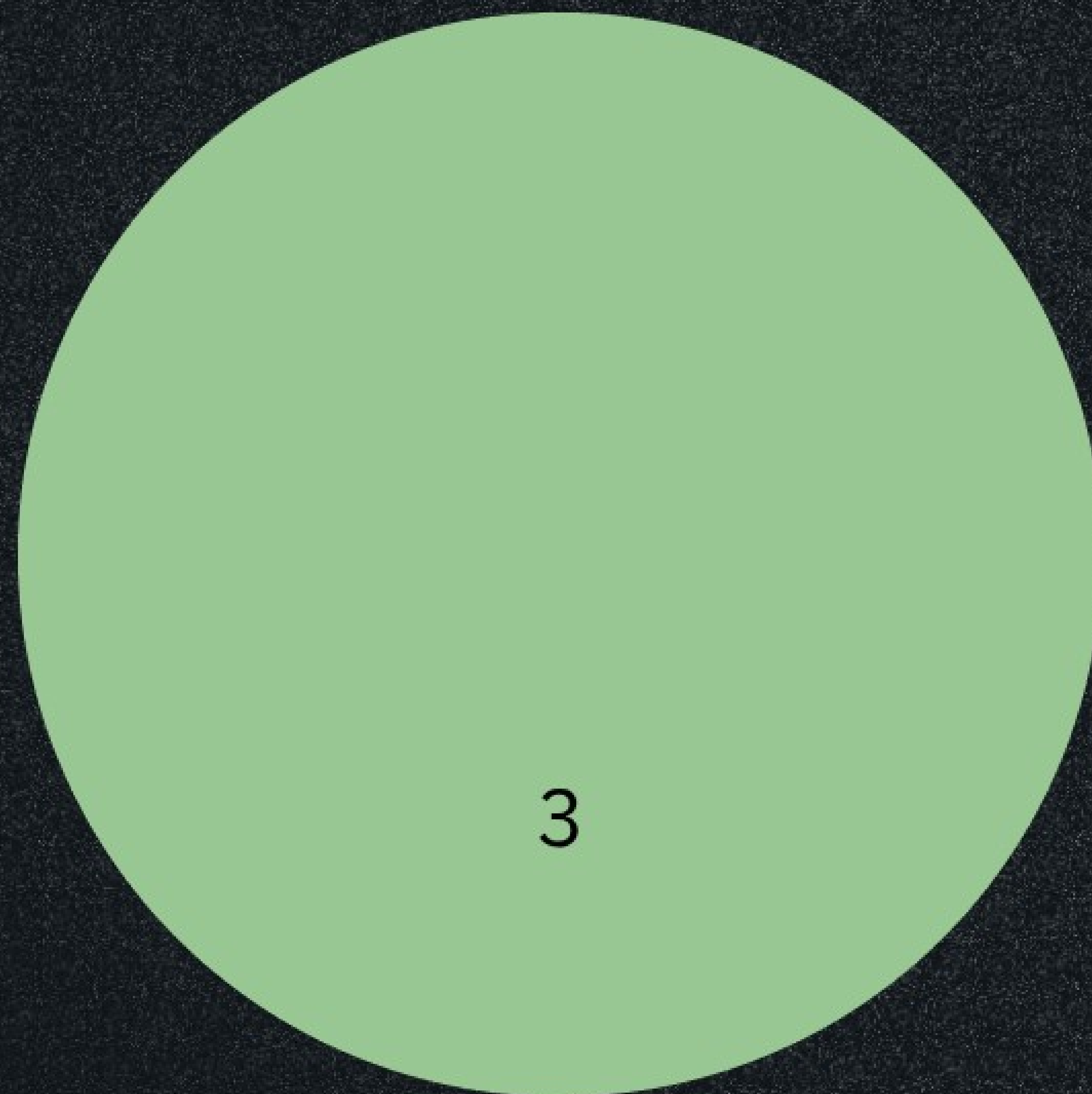
1. If you only have a couple of algorithms and they rarely change, there's no real reason to overcomplicate the code.
2. Clients must be aware of the differences between strategies to be able to select a proper one.
3. Today you can use decrative approach to avoid adding classes and setting structures from them.

What is the primary purpose of the Strategy design pattern?



- | | | |
|---|---|---|
| 2 | A) To encapsulate algorithms, making them interchangeable | ✓ |
| 0 | B) To create objects dynamically | ✗ |
| 0 | C) To set relations between objects | ✗ |
| 0 | D) None of above | ✗ |

Which of the following is a fundamental component of the Strategy pattern?



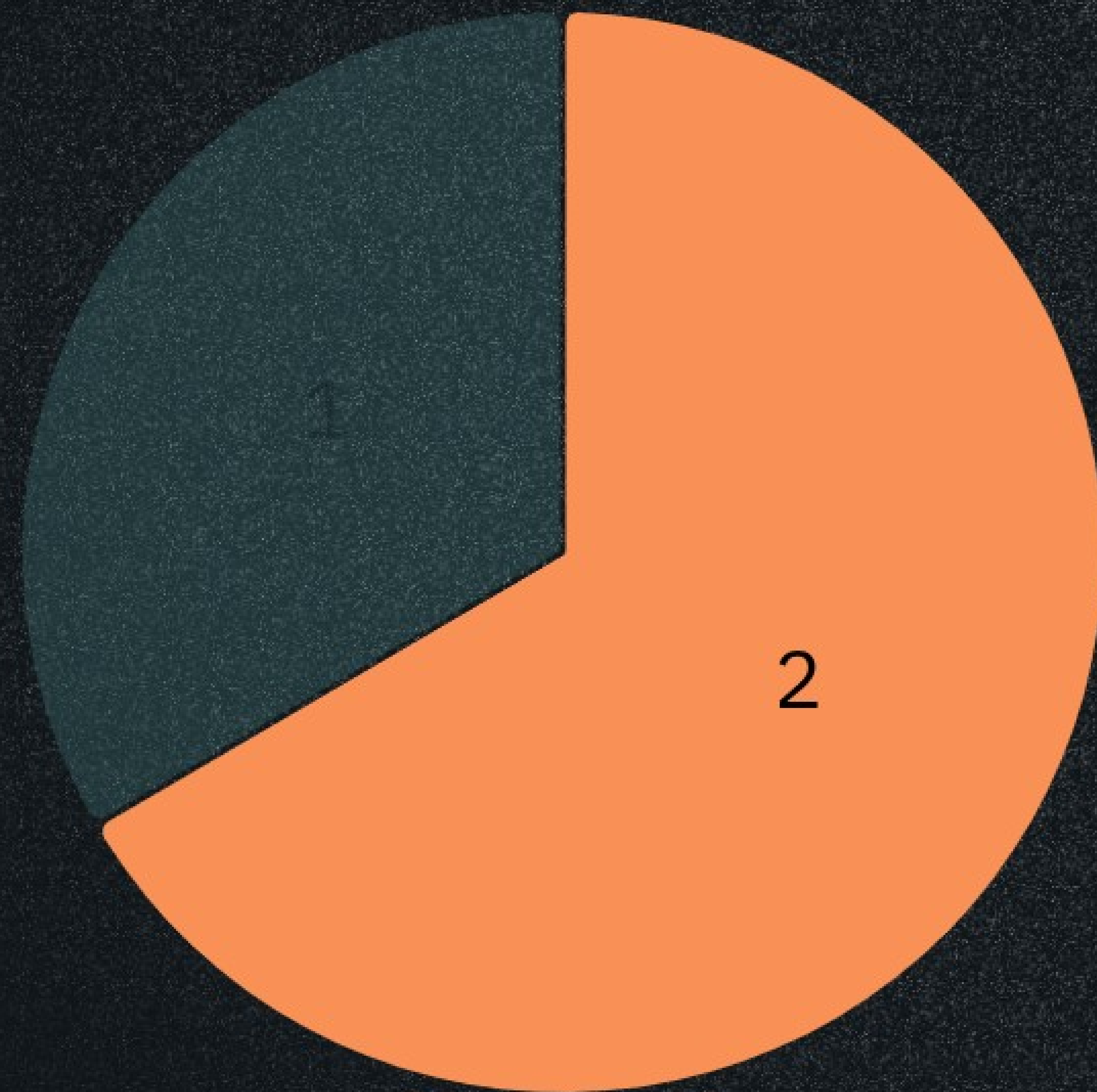
- | | | |
|---|---------------------|---|
| 0 | A) Factory | × |
| 0 | B) Singleton object | × |
| 3 | C) Context | ✓ |
| 0 | D) Mediator | × |

How does the Strategy pattern contribute to a software application's maintainability?



- | | | |
|---|---|---|
| 4 | A) By allowing the addition of new behaviors without modifying existing system code | ✓ |
| 0 | B) By providing a means to store object states | ✗ |
| 0 | C) By managing several databases simultaneously | ✗ |
| 0 | D) By keeping all algorithms in a single class | ✗ |

In the Strategy pattern, what role does the 'Strategy' class play?



2

A) It defines an interface common to all supported strategies



0

B) It maintains a reference to a Strategy object



0

C) It integrates different databases

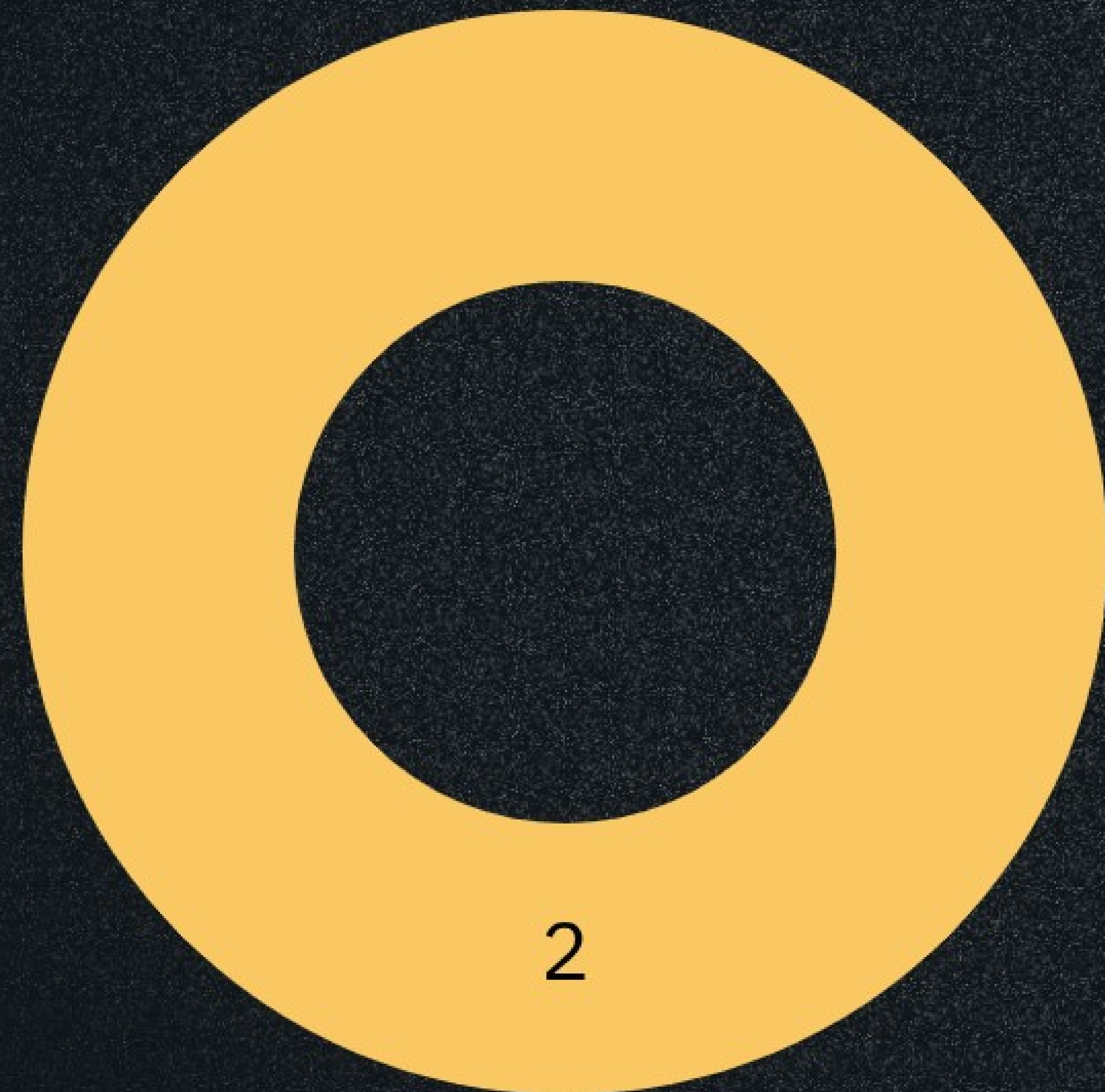


1

D) It ensures there is only one instance of an object throughout the system



When applying the Strategy pattern, how would you allow a program to change its behavior at runtime?



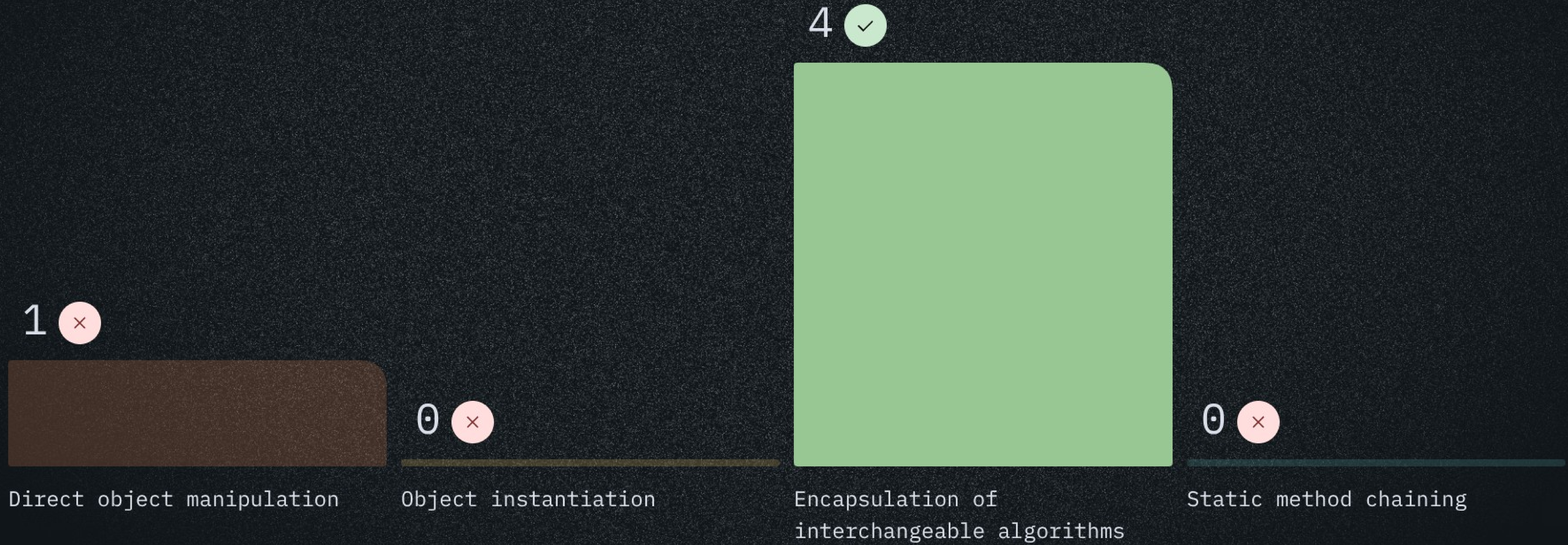
- ☐ 0 A) By modifying the algorithms used directly in the code ☐
- ☒ 2 B) By utilizing polymorphism to change behaviors dynamically using a Strategy interface ☒
- ☐ 0 C) By using a central mediator to control changes ☐
- ☐ 0 D) By creating new objects for each change in behavior ☐

Exam

- Exam Format: same as before - a test on learned materials
- Exam Rules: questions = 26 | time = 45 | max_score = 26
- Materials to prepare: passed modules at learning platform
- Q&A: reach out if you still have any sort of concern

Quiz

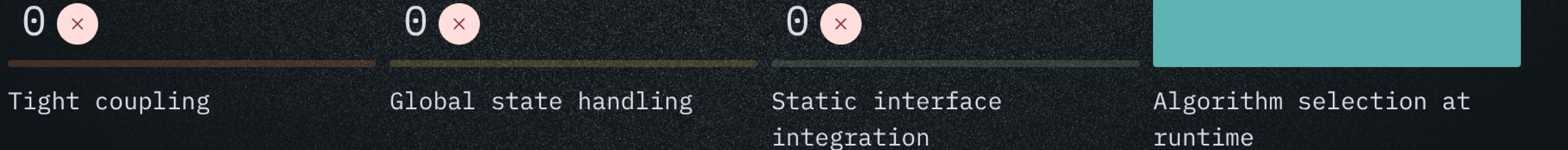
What does the Strategy pattern achieve?



What principle is followed by Strategy?



Which is a key feature of Strategy?



When should Strategy pattern be used?



What does the Strategy pattern promote?

5 ✓



Loose coupling and
behavior encapsulation

0 ✗

Complex inheritance

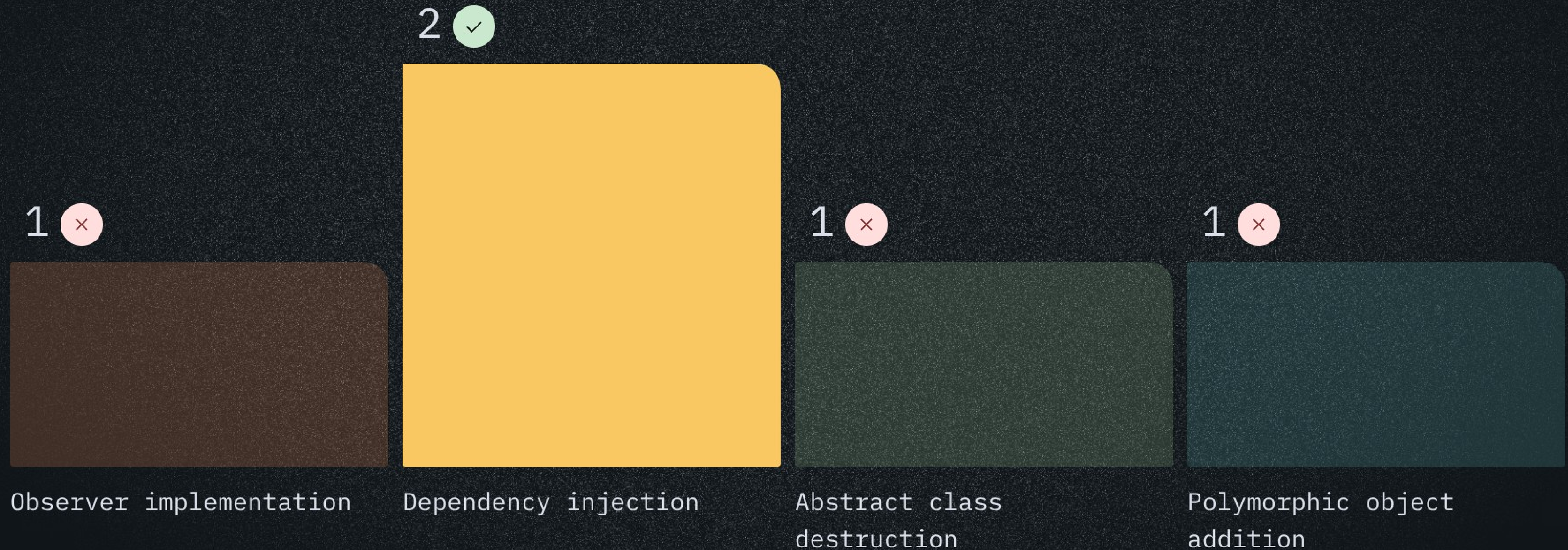
0 ✗

Single class
optimization

0 ✗

Object serialization

Which principle does Strategy align with?



Strategy pattern relies on?

5 ✓



Encapsulation and
interfaces**

0 ✗

Direct class
constructors

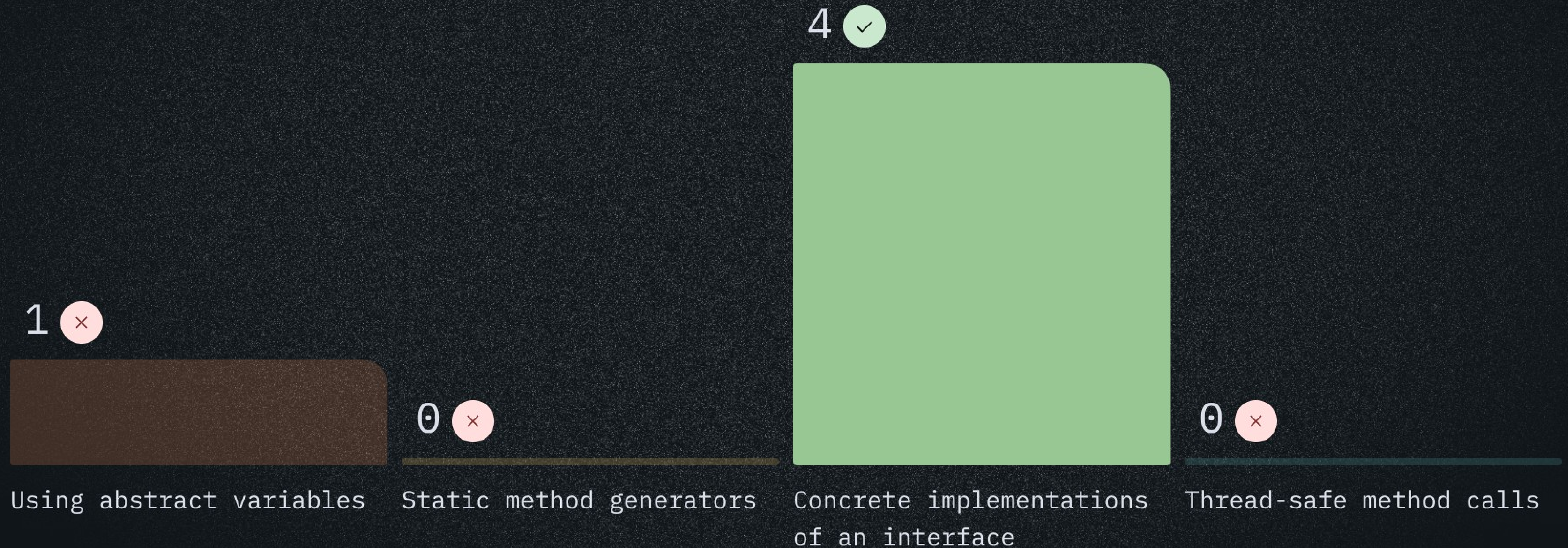
0 ✗

Single inheritance
hierarchy

0 ✗

Global variables

How are behaviors defined in Strategy?



What does the context class do in Strategy?

5 ✓



0 ✗

Delegates behavior
execution to strategies

Add behaviors to
subjects

0 ✗

Directly implements
behaviors

0 ✗

Invokes multiple thread
executions

What happens when you add a new algorithm?

1 ✗



Requires restarting
objects

2 ✗



Changes subject observer
hierarchy

0 ✗



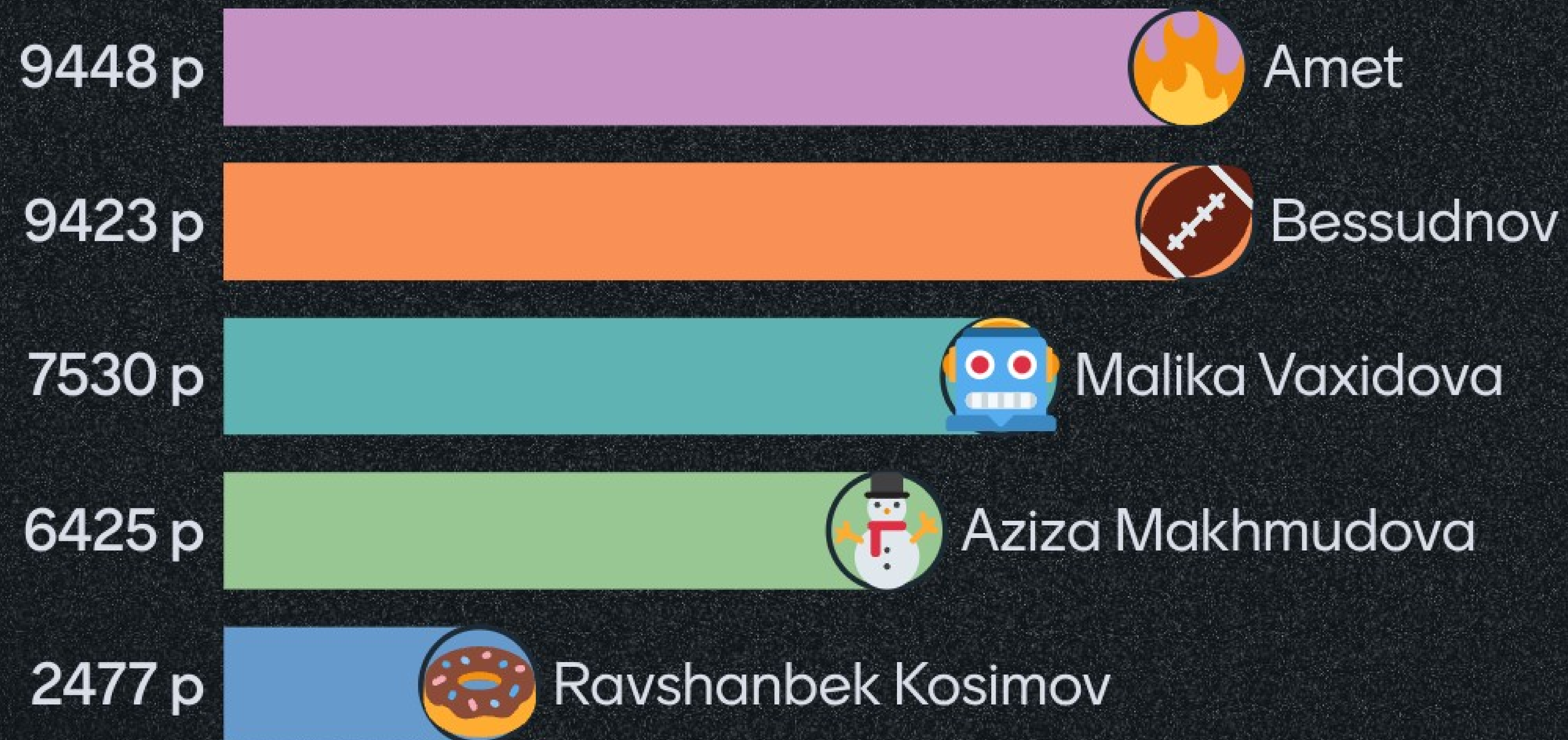
Improves method
serialization

2 ✓



The strategy context
remains unchanged

Quiz leaderboard



Q&A part

0 questions

0 upvotes